Plane Geometry

Section 1: Angles & Parallel Lines

1 Two parallel lines are intersected by a transversal. If one interior angle measures $(3x + 15)^\circ$ and its consecutive interior angle measures $(2x + 45)^\circ$, what is the value of x?

a. 24

b. 30

c. 36

d. 40

2 In a triangle, the ratio of the three interior angles is $2 : 3 : 4$. What is the degree measure of the largest angle?

a. 60°

b. 75°

c. 80°

d. 90°

3 What is the measure of each interior angle of a regular octagon?

a. 120°

b. 135°

c. 140°

d. 144°

Section 2: Triangles, Similarity & Pythagorean Theorem

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4 In a right triangle, the hypotenuse has length 25 cm and one leg has length 7 cm. What is the length of the other leg?

a. 20 cm

b. 24 cm

c. 22 cm

d. 18 cm

5 Two similar triangles have corresponding side lengths in the ratio $3:5$. If the area of the smaller triangle is 36 square units, what is the area of the larger triangle?

a. 60

b. 80

c. 100

d. 125

6 In a 30°-60°-90° right triangle, the length of the hypotenuse is $16\sqrt{3}$. What is the length of the side opposite the 60° angle?

a. 12

b. 24

c. $8\sqrt{3}$

d. 16

Section 3: Quadrilaterals & Polygons

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7 A trapezoid has parallel bases of lengths 14 cm and 26 cm and an altitude of 9 cm. What is the area of the trapezoid?

a. $160\ \text{cm}^2$

b. $180\ \text{cm}^2$

c. $200\ \text{cm}^2$

d. $220\ \text{cm}^2$

8 The diagonals of a rhombus have lengths 16 inches and 30 inches. What is the perimeter of the rhombus?

a. 60 inches

b. 64 inches

c. 68 inches

d. 72 inches

9 A rectangle has a perimeter of 48 meters and a length that is twice its width. What is the area of the rectangle?

a. $96\ \text{m}^2$

b. $128\ \text{m}^2$

c. $144\ \text{m}^2$

d. $160\ \text{m}^2$

Section 4: Circles, Arcs & Sectors

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10 A circle has a radius of 12 cm. What is the area of a sector formed by a central angle of 60°?

a. 12π cm^2

b. 24π cm^2

c. 36π cm^2

d. 48π cm^2

11 In a circle of radius 10 inches, what is the length of an arc intercepted by a central angle of $\frac{3\pi}{5}$ radians?

a. 4π inches

b. 6π inches

c. 8π inches

d. 10π inches

12 An inscribed angle intercepts an arc of 110° on a circle. What is the degree measure of the inscribed angle?

a. 55°

b. 70°

c. 110°

d. 220°

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